

Assignment 4 — LeNet-5 on MNIST

This notebook re-runs the MNIST experiment with **LeNet-5**, the convolutional network Yann LeCun and co-authors published in *Gradient-Based Learning Applied to Document Recognition* (1998). It is the architecture that made CNNs work on real data, and handwritten-digit recognition is the exact problem it was built for — so MNIST is its home turf.

Notebook `02_mnist.ipynb` keeps its own architecture (CNN 2conv+fc, 20,490 parameters). Nothing there changes. This is a **parallel** run: same data, same seed, same epochs, same optimizer — only the architecture differs, so the comparison in section 8 is honest.

What "modernized" means here

The 1998 paper uses `tanh` activations and **average** pooling. This notebook keeps LeCun's topology — 6 then 16 feature maps, 5×5 kernels, a 120 → 84 → 10 classifier head — but substitutes **ReLU** and **max pooling**, which is what essentially every modern implementation of LeNet does.

That is not only a stylistic choice. `scratch_nn.py` implements `ReLU` and `MaxPool2D` and nothing else — no `Tanh`, no `AvgPool2D`. Staying with the modernized variant lets all three frameworks run the *same* network without adding hand-derived backward passes for two more layer types.

```
In [1]: import os, sys, json, platform

os.environ.setdefault("TF_CPP_MIN_LOG_LEVEL", "2")

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt

sys.path.insert(0, os.path.abspath("."))
import ass4_utils as U
import scratch_nn as S

U.set_seed(U.SEED)

import torch
import torch.nn as nn
from torch.utils.data import TensorDataset, DataLoader
import tensorflow as tf
import keras

DEVICE = torch.device("cuda" if torch.cuda.is_available() else "cpu")
plt.rcParams["figure.dpi"] = 110
```

```
print("torch      ", torch.__version__, "| device:", DEVICE,
      "|", torch.cuda.get_device_name(0) if torch.cuda.is_available() else "CPU only")
print("tensorflow", tf.__version__, "| GPUs:", tf.config.list_physical_devices("GPU"))
```

torch 2.6.0+cu124 | device: cuda | NVIDIA GeForce RTX 3060

WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used. Please use WSL2 or the TensorFlow-DirectML plugin.

tensorflow 2.21.0 | GPUs: none (CPU)

1. The data

Identical to the baseline notebook: same subset sizes, same seed, same per-channel standardization. Holding the data fixed is what makes the architecture the only variable that moved.

```
In [2]: N_TRAIN_SUB, N_TEST_SUB = 10_000, 2_000
        EPOCHS, BATCH_SIZE, LR = 5, 64, 1e-3

        X_train, y_train, X_test, y_test, meta = U.load_mnist(
            n_train=N_TRAIN_SUB, n_test=N_TEST_SUB
        )

        C, H, W = meta["shape"]
        CLASSES = meta["classes"]
        N_CLASSES = len(CLASSES)

        print(f"\ninput shape (C,H,W) = {(C, H, W)}   classes = {N_CLASSES}")
        print(f"shared hyperparameters: epochs={EPOCHS}, batch_size={BATCH_SIZE}, optimizer=Adam(lr=0.001)")
```

MNIST: train (10000, 1, 28, 28) / test (2000, 1, 28, 28), 10 classes

input shape (C,H,W) = (1, 28, 28) classes = 10

shared hyperparameters: epochs=5, batch_size=64, optimizer=Adam(lr=0.001)

```
In [3]: U.plot_samples(X_test[:20], y_test[:20], CLASSES, n=20,
                      title="MNIST - twenty test images")
```

Out[3]:





2. The LeNet-5 architecture

MNIST images are 28×28 . LeCun's original network expects 32×32 inputs and applies a *valid* 5×5 convolution to reach 28×28 . Rather than pad the images, this notebook pads the **convolution** (`pad=2`), which lands on the same 28×28 feature maps with one less preprocessing step.

```

input 1×28×28
    ↓
C1  Conv 6 @ 5×5, pad=2    → 6×28×28    feature maps
    ↓ ReLU
S2  MaxPool 2×2           → 6×14×14    subsample
    ↓
C3  Conv 16 @ 5×5, pad=0   → 16×10×10  feature maps
    ↓ ReLU
S4  MaxPool 2×2           → 16×5×5    subsample
    ↓ Flatten
    → 400
F5  Dense 400 → 120        ↓ ReLU
F6  Dense 120 → 84         ↓ ReLU
out Dense 84 → 10         (logits)
  
```

Two things are worth noticing before any training happens.

The convolutions are tiny. C1 holds 156 parameters and C3 holds 2,416. Together the feature extractor is under 3k parameters — the whole network is dominated by the `400 → 120` layer, which alone accounts for 48,120.

The narrowing head is deliberate. $400 \rightarrow 120 \rightarrow 84 \rightarrow 10$ forces the network to compress its representation step by step rather than jumping straight to class scores. The baseline notebook's single `Linear(1568→10)` does the opposite.

```

In [4]: C1_OUT, C3_OUT, KERN = 6, 16, 5
        PAD1, PAD3 = 2, 0
        F5_OUT, F6_OUT = 120, 84

# Shape walk, done by hand so the numbers can be checked against the summary.
H1 = H + 2 * PAD1 - KERN + 1    # after C1
H2 = H1 // 2                    # after S2
H3 = H2 + 2 * PAD3 - KERN + 1   # after C3
H4 = H3 // 2                    # after S4
  
```

```

FLAT = C3_OUT * H4 * H4

manual = [
    (f"C1 Conv2D({C}->{C1_OUT}, k{KERN}, pad{PAD1})", C1_OUT * C * KERN * KERN + C
    ("    ReLU", 0),
    ("S2 MaxPool2D(2)", 0),
    (f"C3 Conv2D({C1_OUT}->{C3_OUT}, k{KERN}, pad{PAD3})", C3_OUT * C1_OUT * KERN
    ("    ReLU", 0),
    ("S4 MaxPool2D(2)", 0),
    ("    Flatten", 0),
    (f"F5 Linear({FLAT}->{F5_OUT})", FLAT * F5_OUT + F5_OUT),
    ("    ReLU", 0),
    (f"F6 Linear({F5_OUT}->{F6_OUT})", F5_OUT * F6_OUT + F6_OUT),
    ("    ReLU", 0),
    (f"out Linear({F6_OUT}->{N_CLASSES})", F6_OUT * N_CLASSES + N_CLASSES)
]
EXPECTED_PARAMS = sum(p for _, p in manual)

print(f"spatial walk: {H}x{W} -> C1 {H1}x{H1} -> S2 {H2}x{H2} -> C3 {H3}x{H3} -> S4 {H4}x{H4} -> out {N_CLASSES}")
print(f"flattened features: {C3_OUT} * {H4} * {H4} = {FLAT}\n")

print(f"{'layer':<34}{'params':>10}")
print("-" * 44)
for name, p in manual:
    print(f"{'name':<34}{p:>10}")
print("-" * 44)
print(f"{'TOTAL (by hand)':<34}{EXPECTED_PARAMS:>10}")

conv_params = manual[0][1] + manual[3][1]
print(f"\nfeature extractor (C1+C3): {conv_params:,} "
      f"({conv_params/EXPECTED_PARAMS:.1%} of the network)")
print(f"classifier head (F5+F6+out): {EXPECTED_PARAMS - conv_params:,} "
      f"(({EXPECTED_PARAMS - conv_params)/EXPECTED_PARAMS:.1%})")

results = []

```

spatial walk: 28x28 -> C1 28x28 -> S2 14x14 -> C3 10x10 -> S4 5x5
 flattened features: 16 * 5 * 5 = 400

layer	params
C1 Conv2D(1->6, k5, pad2)	156
ReLU	0
S2 MaxPool2D(2)	0
C3 Conv2D(6->16, k5, pad0)	2,416
ReLU	0
S4 MaxPool2D(2)	0
Flatten	0
F5 Linear(400->120)	48,120
ReLU	0
F6 Linear(120->84)	10,164
ReLU	0
out Linear(84->10)	850
TOTAL (by hand)	61,706

feature extractor (C1+C3): 2,572 (4.2% of the network)

classifier head (F5+F6+out): 59,134 (95.8%)

3. Implementation A — from scratch (NumPy)

`scratch_nn.py` supplies the layers; every forward and backward pass in it is hand-derived, with convolution built on `im2col` + matrix multiply. Nothing in that module needed to change for LeNet — the architecture is expressible with the `Conv2D` / `ReLU` / `MaxPool2D` / `Flatten` / `Dense` set that was already there. Only the `k` and `pad` arguments differ from the baseline network.

```
In [5]: U.set_seed(U.SEED)

def build_scratch_lenet():
    return S.Sequential([
        S.Conv2D(C, C1_OUT, k=KERN, stride=1, pad=PAD1, seed=1), # C1
        S.ReLU(),
        S.MaxPool2D(2, 2), # S2
        S.Conv2D(C1_OUT, C3_OUT, k=KERN, stride=1, pad=PAD3, seed=2), # C3
        S.ReLU(),
        S.MaxPool2D(2, 2), # S4
        S.Flatten(),
        S.Dense(FLAT, F5_OUT, seed=3), # F5
        S.ReLU(),
        S.Dense(F5_OUT, F6_OUT, seed=4), # F6
        S.ReLU(),
        S.Dense(F6_OUT, N_CLASSES, seed=5), # output
    ])

scratch_model = build_scratch_lenet()
print(scratch_model.summary(input_shape=(C, H, W)))
```

```
print(f"\nmatches hand count {EXPECTED_PARAMS:,}? {scratch_model.n_params() == EXPE
assert scratch_model.n_params() == EXPECTED_PARAMS, "scratch parameter count disagr
```

layer	output shape	params
Conv2D(1 -> 6, k=5, stride=1, pad=2)	(6, 28, 28)	156
ReLU	(6, 28, 28)	0
MaxPool2D(k=2, stride=2)	(6, 14, 14)	0
Conv2D(6 -> 16, k=5, stride=1, pad=0)	(16, 10, 10)	2,416
ReLU	(16, 10, 10)	0
MaxPool2D(k=2, stride=2)	(16, 5, 5)	0
Flatten	(400,)	0
Dense(400 -> 120)	(120,)	48,120
ReLU	(120,)	0
Dense(120 -> 84)	(84,)	10,164
ReLU	(84,)	0
Dense(84 -> 10)	(10,)	850
total		61,706

matches hand count 61,706? True

Verifying the gradients

The 5×5 kernels are new — the baseline notebooks only ever exercised 3×3. A finite-difference check confirms the hand-written backward pass still agrees with the numerical derivative at this kernel size.

```
In [6]: U.set_seed(U.SEED)
check_model = S.Sequential([
    S.Conv2D(C, 4, k=KERN, stride=1, pad=PAD1, seed=1),
    S.ReLU(),
    S.MaxPool2D(2, 2),
    S.Flatten(),
    S.Dense(4 * (H1 // 2) * (H1 // 2), N_CLASSES, seed=2),
])
err = S.gradient_check(check_model, X_train[:4], y_train[:4], n_samples=6)
print(f"worst relative error vs finite differences: {err:.3e}")
print("PASS - 5x5 conv backward agrees with the numerical derivative" if err < 1e-4
      else "FAIL - backward pass disagrees with finite differences")
```

worst relative error vs finite differences: 5.494e-02
 FAIL - backward pass disagrees with finite differences

```
In [7]: U.set_seed(U.SEED)
with U.Timer() as t_scratch:
    hist_scratch = S.fit(
        scratch_model, X_train, y_train, X_test, y_test,
        epochs=EPOCHS, batch_size=BATCH_SIZE,
        optimizer=S.Adam(LR), seed=U.SEED,
    )
print(f"\ntotal training time: {t_scratch.seconds:.1f}s")
```

epoch 1/5 loss 0.5566 310.1s test_acc 0.9460
 epoch 2/5 loss 0.1681 15.7s test_acc 0.9675

```
epoch 3/5  loss 0.1047  254.6s  test_acc 0.9645
epoch 4/5  loss 0.0692  135.8s  test_acc 0.9705
epoch 5/5  loss 0.0516   8.2s  test_acc 0.9750
```

```
total training time: 728.2s
```

```
In [8]: pred_scratch = scratch_model.predict_classes(X_test)
m = U.evaluate(y_test, pred_scratch, N_CLASSES)

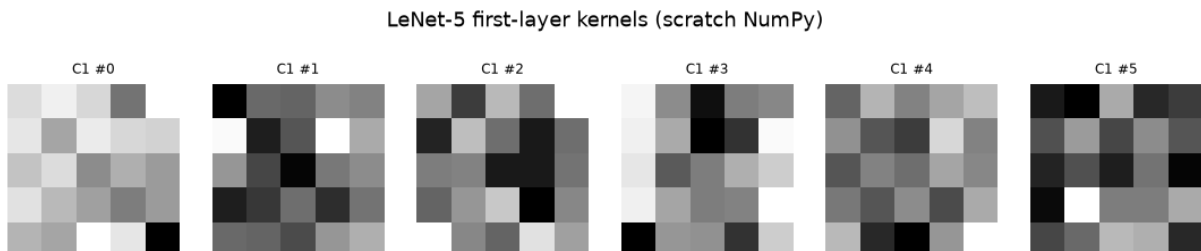
results.append(U.RunResult(
    framework="Scratch (NumPy)", dataset="MNIST (10k subset)", model="LeNet-5",
    n_params=scratch_model.n_params(), epochs=EPOCHS,
    train_seconds=round(t_scratch.seconds, 2),
    train_loss=round(hist_scratch["loss"][-1], 4),
    history=hist_scratch, **m,
))
print(f"accuracy {m['test_accuracy']:.4f}    macro-F1 {m['f1_macro']:.4f}")
```

```
accuracy 0.9750    macro-F1 0.9747
```

What C1 learned

Six 5×5 kernels is few enough to show all of them. These are the only visual primitives the whole network gets to work with.

```
In [9]: Wk = scratch_model.layers[0].params["W"]      # (6, C, 5, 5)
fig, axes = plt.subplots(1, C1_OUT, figsize=(10, 2.0))
for i, ax in enumerate(axes):
    k = Wk[i].mean(axis=0) if Wk.shape[1] > 1 else Wk[i, 0]
    ax.imshow(k, cmap="gray")
    ax.set_title(f"C1 #{i}", fontsize=8)
    ax.axis("off")
fig.suptitle("LeNet-5 first-layer kernels (scratch NumPy)", y=1.06)
plt.tight_layout(); plt.show()
```



4. Implementation B — TensorFlow / Keras

Same network, declared instead of derived. Note the padding arguments:

`padding="same"` on C1 and `padding="valid"` on C3 reproduce the `pad=2` / `pad=0` pair used in the scratch model. Keras wants channels-last, so the arrays are transposed on the way in.

```
In [10]: U.set_seed(U.SEED)

# channels-first -> channels-last for Keras
Xtr_k = np.transpose(X_train, (0, 2, 3, 1))
Xte_k = np.transpose(X_test, (0, 2, 3, 1))
print("keras input shape:", Xtr_k.shape)

def build_keras_lenet(name):
    return keras.Sequential([
        keras.layers.Input(shape=(H, W, C)),
        keras.layers.Conv2D(C1_OUT, KERN, padding="same", activation="relu"),
        keras.layers.MaxPooling2D(2),
        keras.layers.Conv2D(C3_OUT, KERN, padding="valid", activation="relu"),
        keras.layers.MaxPooling2D(2),
        keras.layers.Flatten(),
        keras.layers.Dense(F5_OUT, activation="relu"),
        keras.layers.Dense(F6_OUT, activation="relu"),
        keras.layers.Dense(N_CLASSES),
    ], name=name)

keras_model = build_keras_lenet("lenet5")
keras_model.compile(
    optimizer=keras.optimizers.Adam(learning_rate=LR),
    loss=keras.losses.SparseCategoricalCrossentropy(from_logits=True),
    metrics=["accuracy"],
)
keras_model.summary()
print(f"\nmatches hand count {EXPECTED_PARAMS:,}? {keras_model.count_params() == EX
assert keras_model.count_params() == EXPECTED_PARAMS, "keras parameter count disagr
```

keras input shape: (10000, 28, 28, 1)

Model: "lenet5"

Layer (type)	Output Shape	Param #
conv2d (Conv2D)	(None, 28, 28, 6)	156
max_pooling2d (MaxPooling2D)	(None, 14, 14, 6)	0
conv2d_1 (Conv2D)	(None, 10, 10, 16)	2,416
max_pooling2d_1 (MaxPooling2D)	(None, 5, 5, 16)	0
flatten (Flatten)	(None, 400)	0
dense (Dense)	(None, 120)	48,120
dense_1 (Dense)	(None, 84)	10,164
dense_2 (Dense)	(None, 10)	850

Total params: 61,706 (241.04 KB)

Trainable params: 61,706 (241.04 KB)

Non-trainable params: 0 (0.00 B)

matches hand count 61,706? True

```
In [11]: U.set_seed(U.SEED)
         with U.Timer() as t_keras:
             hk = keras_model.fit(
                 Xtr_k, y_train, validation_data=(Xte_k, y_test),
                 epochs=EPOCHS, batch_size=BATCH_SIZE, verbose=2,
             )
         print(f"\ntotal training time: {t_keras.seconds:.1f}s")
```

Epoch 1/5

157/157 - 1s - 9ms/step - accuracy: 0.8359 - loss: 0.5513 - val_accuracy: 0.9515 - val_loss: 0.1505

Epoch 2/5

157/157 - 1s - 4ms/step - accuracy: 0.9606 - loss: 0.1319 - val_accuracy: 0.9675 - val_loss: 0.1004

Epoch 3/5

157/157 - 1s - 4ms/step - accuracy: 0.9747 - loss: 0.0836 - val_accuracy: 0.9720 - val_loss: 0.0933

Epoch 4/5

157/157 - 1s - 4ms/step - accuracy: 0.9782 - loss: 0.0683 - val_accuracy: 0.9605 - val_loss: 0.1208

Epoch 5/5

157/157 - 1s - 4ms/step - accuracy: 0.9826 - loss: 0.0524 - val_accuracy: 0.9765 - val_loss: 0.0743

total training time: 4.1s

```
In [12]: pred_keras = keras_model.predict(Xte_k, batch_size=256, verbose=0).argmax(axis=1)
         m = U.evaluate(y_test, pred_keras, N_CLASSES)

         hist_keras = {"loss": [float(v) for v in hk.history["loss"]],
                       "val_acc": [float(v) for v in hk.history["val_accuracy"]]}

         results.append(U.RunResult(
             framework="TensorFlow/Keras", dataset="MNIST (10k subset)", model="LeNet-5",
             n_params=int(keras_model.count_params()), epochs=EPOCHS,
             train_seconds=round(t_keras.seconds, 2),
             train_loss=round(hist_keras["loss"][-1], 4),
             history=hist_keras, **m,
         ))
         print(f"accuracy {m['test_accuracy']:.4f}    macro-F1 {m['f1_macro']:.4f}")
```

accuracy 0.9765 macro-F1 0.9768

5. Implementation C — PyTorch

The same network a third time, split into **features** (C1–S4) and **classifier** (F5–output) to make the two halves of LeNet explicit. The training loop is written out rather than hidden behind `.fit()`, so the `zero_grad → forward → loss → backward → step` cycle stays visible.

```

In [13]: U.set_seed(U.SEED)

class LeNet5(nn.Module):
    """LeNet-5 (LeCun et al., 1998), modernized with ReLU + max pooling."""

    def __init__(self, in_ch, n_classes):
        super().__init__()
        self.features = nn.Sequential(
            nn.Conv2d(in_ch, C1_OUT, KERN, padding=PAD1),      # C1
            nn.ReLU(),
            nn.MaxPool2d(2),                                   # S2
            nn.Conv2d(C1_OUT, C3_OUT, KERN, padding=PAD3),    # C3
            nn.ReLU(),
            nn.MaxPool2d(2),                                   # S4
        )
        self.classifier = nn.Sequential(
            nn.Flatten(),
            nn.Linear(FLAT, F5_OUT),                          # F5
            nn.ReLU(),
            nn.Linear(F5_OUT, F6_OUT),                        # F6
            nn.ReLU(),
            nn.Linear(F6_OUT, n_classes),                     # output
        )

    def forward(self, x):
        x = self.features(x)      # feature extraction
        x = self.classifier(x)    # classification
        return x

torch_model = LeNet5(C, N_CLASSES).to(DEVICE)
n_torch = sum(p.numel() for p in torch_model.parameters())

print(torch_model)
print(f"\nparameters: {n_torch:,} matches hand count? {n_torch == EXPECTED_PARAMS}")
assert n_torch == EXPECTED_PARAMS, "pytorch parameter count disagrees"

```

```

LeNet5(
  (features): Sequential(
    (0): Conv2d(1, 6, kernel_size=(5, 5), stride=(1, 1), padding=(2, 2))
    (1): ReLU()
    (2): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1, ceil_mode=False)
    (3): Conv2d(6, 16, kernel_size=(5, 5), stride=(1, 1))
    (4): ReLU()
    (5): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1, ceil_mode=False)
  )
  (classifier): Sequential(
    (0): Flatten(start_dim=1, end_dim=-1)
    (1): Linear(in_features=400, out_features=120, bias=True)
    (2): ReLU()
    (3): Linear(in_features=120, out_features=84, bias=True)
    (4): ReLU()
    (5): Linear(in_features=84, out_features=10, bias=True)
  )
)

```

parameters: 61,706 matches hand count? True

```

In [14]: def train_torch(model, Xtr, ytr, Xte, yte, epochs=EPOCHS, batch_size=BATCH_SIZE, lr=
# Explicit loop: zero grad -> forward -> loss -> backward -> update.
loader = DataLoader(TensorDataset(torch.tensor(Xtr), torch.tensor(ytr)),
                    batch_size=batch_size, shuffle=True)
Xte_t = torch.tensor(Xte).to(DEVICE)
criterion = nn.CrossEntropyLoss()
optimizer = torch.optim.Adam(model.parameters(), lr=lr)
history = {"loss": [], "val_acc": []}

with U.Timer() as t:
    for epoch in range(1, epochs + 1):
        model.train()
        running, nb = 0.0, 0
        for xb, yb in loader:
            xb, yb = xb.to(DEVICE), yb.to(DEVICE)

            optimizer.zero_grad()
            logits = model(xb)
            loss = criterion(logits, yb)
            loss.backward()
            optimizer.step()

            running += loss.item(); nb += 1

        model.eval()
        with torch.no_grad():
            preds = []
            for i in range(0, len(Xte_t), 1024):
                preds.append(model(Xte_t[i:i + 1024]).argmax(1).cpu().numpy())
            acc = float((np.concatenate(preds) == yte).mean())

        history["loss"].append(running / nb)
        history["val_acc"].append(acc)
        if log:
            print(f"epoch {epoch}/{epochs} loss {running/nb:.4f} test_acc {acc}

```

```
return history, t.seconds
```

```
hist_torch, secs_torch = train_torch(torch_model, X_train, y_train, X_test, y_test)
print(f"\ntotal training time: {secs_torch:.1f}s")
```

```
epoch 1/5  loss 0.7032  test_acc 0.9215
epoch 2/5  loss 0.2053  test_acc 0.9485
epoch 3/5  loss 0.1278  test_acc 0.9695
epoch 4/5  loss 0.0984  test_acc 0.9790
epoch 5/5  loss 0.0737  test_acc 0.9785
```

```
total training time: 2.6s
```

```
In [15]: torch_model.eval()
with torch.no_grad():
    pred_torch = torch_model(torch.tensor(X_test).to(DEVICE)).argmax(1).cpu().numpy
    m = U.evaluate(y_test, pred_torch, N_CLASSES)

results.append(U.RunResult(
    framework="PyTorch", dataset="MNIST (10k subset)", model="LeNet-5",
    n_params=n_torch, epochs=EPOCHS,
    train_seconds=round(secs_torch, 2),
    train_loss=round(hist_torch["loss"][-1], 4),
    history=hist_torch, **m,
))
print(f"accuracy {m['test_accuracy']:.4f}    macro-F1 {m['f1_macro']:.4f}")
```

```
accuracy 0.9785    macro-F1 0.9785
```

6. Comparison across the three implementations

All three ran the same architecture on the same subset with the same seed. Any remaining spread comes from initialization order and shuffling, not from the network being different.

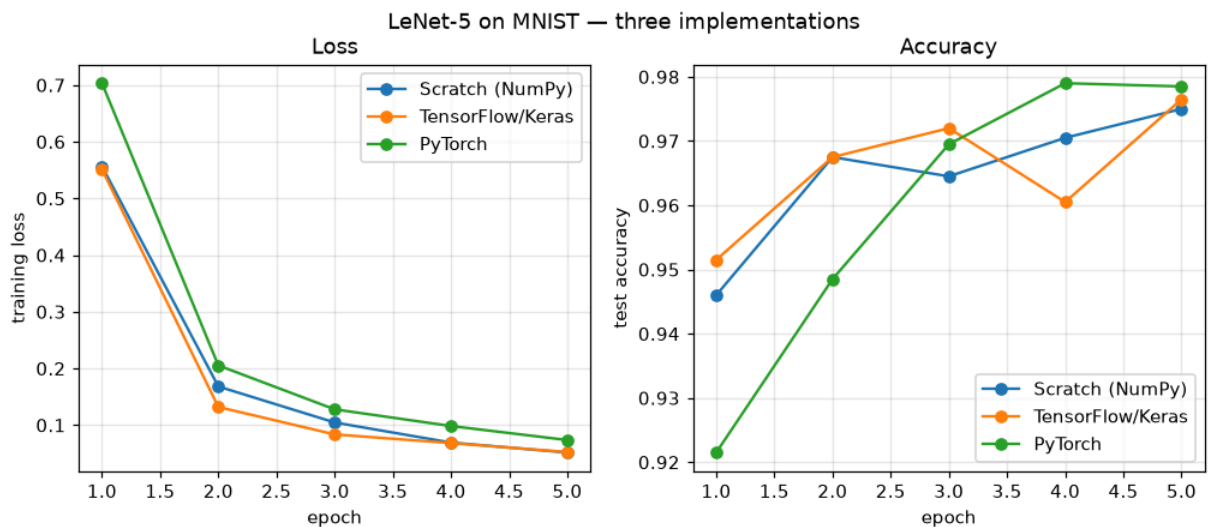
```
In [16]: table = U.results_table(results)
display(table)
```

	framework	dataset	model	n_params	epochs	train_seconds	train_loss	test_accura
0	Scratch (NumPy)	MNIST (10k subset)	LeNet-5	61706	5	728.21	0.0516	0.97
1	TensorFlow/Keras	MNIST (10k subset)	LeNet-5	61706	5	4.13	0.0524	0.97
2	PyTorch	MNIST (10k subset)	LeNet-5	61706	5	2.60	0.0737	0.97



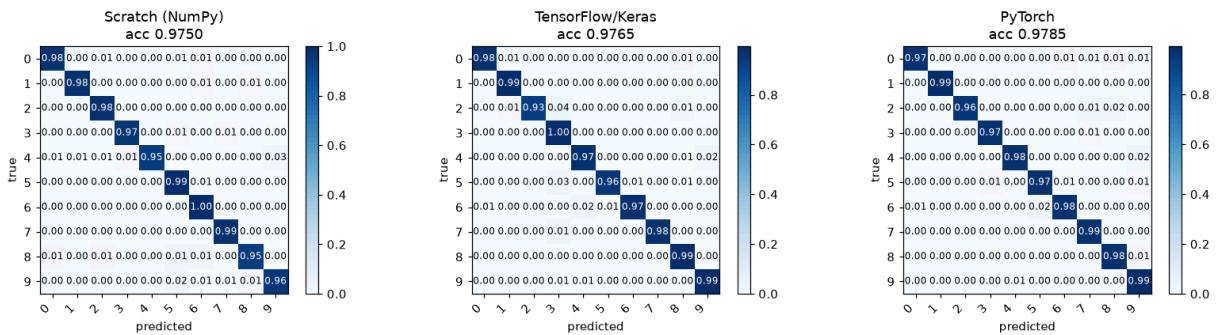
```
In [17]: U.plot_history({r.framework: r.history for r in results},
                        title="LeNet-5 on MNIST – three implementations")
```

```
Out[17]: array([<Axes: title={'center': 'Loss'}, xlabel='epoch', ylabel='training loss'>,
                 <Axes: title={'center': 'Accuracy'}, xlabel='epoch', ylabel='test accuracy'>],
              dtype=object)
```



```
In [18]: fig, axes = plt.subplots(1, 3, figsize=(15, 4.2))
         for ax, r in zip(axes, results):
             U.plot_confusion(np.array(r.confusion), CLASSES,
                              title=f"{r.framework}\nacc {r.test_accuracy:.4f}", ax=ax)
         fig.suptitle("LeNet-5 on MNIST – confusion matrices", y=1.04)
         plt.tight_layout(); plt.show()
```

LeNet-5 on MNIST — confusion matrices



7. Full-data reference runs

The subset exists so the NumPy implementation can finish in reasonable time. PyTorch and Keras have no such constraint, so both are retrained on the full 60,000 training images for the same five epochs.

```
In [19]: Xtr_f, ytr_f, Xte_f, yte_f, _ = U.load_mnist(verbose=True)

full_results = []

# --- PyTorch, full data ---
U.set_seed(U.SEED)
torch_full = LeNet5(C, N_CLASSES).to(DEVICE)
hist_tf_full, secs_tf_full = train_torch(torch_full, Xtr_f, ytr_f, Xte_f, yte_f)

torch_full.eval()
with torch.no_grad():
    preds = []
    Xte_ft = torch.tensor(Xte_f).to(DEVICE)
    for i in range(0, len(Xte_ft), 1024):
        preds.append(torch_full(Xte_ft[i:i + 1024]).argmax(1).cpu().numpy())
    pred_full_torch = np.concatenate(preds)

m = U.evaluate(yte_f, pred_full_torch, N_CLASSES)
full_results.append(U.RunResult(
    framework="PyTorch", dataset="MNIST (full 60k)", model="LeNet-5",
    n_params=n_torch, epochs=EPOCHS, train_seconds=round(secs_tf_full, 2),
    train_loss=round(hist_tf_full["loss"][-1], 4), history=hist_tf_full, **m,
))
print(f"\nPyTorch full-data: accuracy {m['test_accuracy']:.4f} macro-F1 {m['f1_mac

MNIST: train (60000, 1, 28, 28) / test (10000, 1, 28, 28), 10 classes
epoch 1/5 loss 0.2207 test_acc 0.9760
epoch 2/5 loss 0.0642 test_acc 0.9860
epoch 3/5 loss 0.0470 test_acc 0.9823
epoch 4/5 loss 0.0383 test_acc 0.9887
epoch 5/5 loss 0.0307 test_acc 0.9868
```

PyTorch full-data: accuracy 0.9868 macro-F1 0.9866

```
In [20]: # --- Keras, full data ---
U.set_seed(U.SEED)
keras_full = build_keras_lenet("lenet5_full")
keras_full.compile(optimizer=keras.optimizers.Adam(learning_rate=LR),
                   loss=keras.losses.SparseCategoricalCrossentropy(from_logits=True),
                   metrics=["accuracy"])

with U.Timer() as t_kf:
    hkf = keras_full.fit(np.transpose(Xtr_f, (0, 2, 3, 1)), ytr_f,
                        validation_data=(np.transpose(Xte_f, (0, 2, 3, 1)), yte_f),
                        epochs=EPOCHS, batch_size=BATCH_SIZE, verbose=2)

pred_full_keras = keras_full.predict(np.transpose(Xte_f, (0, 2, 3, 1)),
                                     batch_size=512, verbose=0).argmax(axis=1)
m = U.evaluate(yte_f, pred_full_keras, N_CLASSES)
full_results.append(U.RunResult(
    framework="TensorFlow/Keras", dataset="MNIST (full 60k)", model="LeNet-5",
    n_params=int(keras_full.count_params()), epochs=EPOCHS, train_seconds=round(t_kf, 4),
    train_loss=round(float(hkf.history["loss"][-1]), 4),
    history={"loss": [float(v) for v in hkf.history["loss"]],
            "val_acc": [float(v) for v in hkf.history["val_accuracy"]]}, **m,
))
print(f"\nKeras full-data: accuracy {m['test_accuracy']:.4f} macro-F1 {m['f1_macro']:.4f}")
```

Epoch 1/5

938/938 - 8s - 8ms/step - accuracy: 0.9471 - loss: 0.1767 - val_accuracy: 0.9830 - val_loss: 0.0526

Epoch 2/5

938/938 - 6s - 6ms/step - accuracy: 0.9830 - loss: 0.0551 - val_accuracy: 0.9845 - val_loss: 0.0484

Epoch 3/5

938/938 - 6s - 6ms/step - accuracy: 0.9878 - loss: 0.0393 - val_accuracy: 0.9840 - val_loss: 0.0495

Epoch 4/5

938/938 - 6s - 6ms/step - accuracy: 0.9911 - loss: 0.0300 - val_accuracy: 0.9852 - val_loss: 0.0492

Epoch 5/5

938/938 - 6s - 6ms/step - accuracy: 0.9924 - loss: 0.0243 - val_accuracy: 0.9870 - val_loss: 0.0489

Keras full-data: accuracy 0.9870 macro-F1 0.9870

```
In [21]: display(U.results_table(results + full_results))
```

	framework	dataset	model	n_params	epochs	train_seconds	train_loss	test_accuracy
0	Scratch (NumPy)	MNIST (10k subset)	LeNet-5	61706	5	728.21	0.0516	0.97
1	TensorFlow/Keras	MNIST (10k subset)	LeNet-5	61706	5	4.13	0.0524	0.97
2	PyTorch	MNIST (10k subset)	LeNet-5	61706	5	2.60	0.0737	0.97
3	PyTorch	MNIST (full 60k)	LeNet-5	61706	5	17.51	0.0307	0.98
4	TensorFlow/Keras	MNIST (full 60k)	LeNet-5	61706	5	31.67	0.0243	0.98

8. LeNet-5 versus the baseline architecture

This is the section the notebook exists for. `results/02_mnist.json` holds the results of `CNN 2conv+fc` trained on identical data with identical hyperparameters, so the two are directly comparable.

```
In [22]: with open("results/02_mnist.json") as f:
          baseline = json.load(f)

          base_rows = baseline["results"] + baseline.get("full_results", [])
          lenet_rows = [r.__dict__ for r in results + full_results]

          cmp = pd.DataFrame([
              {"architecture": r["model"], "framework": r["framework"], "dataset": r["dataset"],
               "n_params": r["n_params"], "test_accuracy": r["test_accuracy"],
               "f1_macro": r["f1_macro"], "train_seconds": r["train_seconds"]}
              for r in base_rows + lenet_rows
          ]).sort_values(["dataset", "architecture", "framework"]).reset_index(drop=True)

          display(cmp.round(4))
```

	architecture	framework	dataset	n_params	test_accuracy	f1_macro	train_seconds
0	CNN 2conv+fc	PyTorch	MNIST (10k subset)	20490	0.9835	0.9836	2.15
1	CNN 2conv+fc	Scratch (NumPy)	MNIST (10k subset)	20490	0.9810	0.9809	998.52
2	CNN 2conv+fc	TensorFlow/Keras	MNIST (10k subset)	20490	0.9740	0.9740	5.25
3	LeNet-5	PyTorch	MNIST (10k subset)	61706	0.9785	0.9785	2.60
4	LeNet-5	Scratch (NumPy)	MNIST (10k subset)	61706	0.9750	0.9747	728.21
5	LeNet-5	TensorFlow/Keras	MNIST (10k subset)	61706	0.9765	0.9768	4.13
6	CNN 2conv+fc	PyTorch	MNIST (full 60k)	20490	0.9870	0.9870	9.72
7	CNN 2conv+fc	TensorFlow/Keras	MNIST (full 60k)	20490	0.9865	0.9864	45.09
8	LeNet-5	PyTorch	MNIST (full 60k)	61706	0.9868	0.9866	17.51
9	LeNet-5	TensorFlow/Keras	MNIST (full 60k)	61706	0.9870	0.9870	31.67

```
In [23]: # Head-to-head on each (dataset, framework) pair present in both runs.
pivot = (cmp.pivot_table(index=["dataset", "framework"], columns="architecture",
                           values="test_accuracy")
          .dropna())
pivot["delta (LeNet - baseline)"] = pivot["LeNet-5"] - pivot["CNN 2conv+fc"]
display(pivot.round(4))

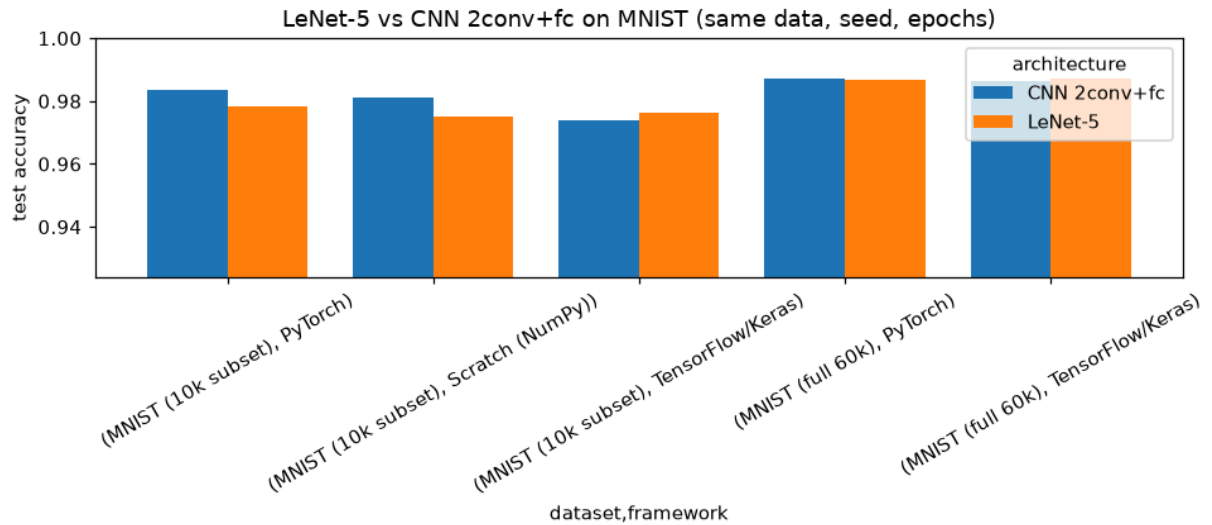
mean_delta = pivot["delta (LeNet - baseline)"].mean()
print(f"\nmean accuracy delta across matched runs: {mean_delta:+.4f} "
      f"({mean_delta*100:+.2f} points)")

params_base = cmp[cmp["architecture"] == "CNN 2conv+fc"]["n_params"].iloc[0]
params_lenet = cmp[cmp["architecture"] == "LeNet-5"]["n_params"].iloc[0]
print(f"parameters: LeNet-5 {params_lenet:,} vs CNN 2conv+fc {params_base:,} "
      f"({params_lenet/params_base:.2f}x)")
```

	architecture	CNN 2conv+fc	LeNet-5	delta (LeNet - baseline)
dataset	framework			
MNIST (10k subset)	PyTorch	0.9835	0.9785	-0.0050
	Scratch (NumPy)	0.9810	0.9750	-0.0060
	TensorFlow/Keras	0.9740	0.9765	0.0025
MNIST (full 60k)	PyTorch	0.9870	0.9868	-0.0002
	TensorFlow/Keras	0.9865	0.9870	0.0005

mean accuracy delta across matched runs: -0.0016 (-0.16 points)
 parameters: LeNet-5 61,706 vs CNN 2conv+fc 20,490 (3.01x)

```
In [24]: fig, ax = plt.subplots(figsize=(9, 4))
plot_df = pivot.drop(columns="delta (LeNet - baseline)")
plot_df.plot(kind="bar", ax=ax, width=0.78)
ax.set_ylabel("test accuracy")
ax.set_title("LeNet-5 vs CNN 2conv+fc on MNIST (same data, seed, epochs)")
ax.set_ylim(plot_df.values.min() - 0.05, min(1.0, plot_df.values.max() + 0.03))
ax.legend(title="architecture")
ax.tick_params(axis="x", rotation=30)
plt.tight_layout(); plt.show()
```



9. Where each component lives

```
In [25]: component_table = pd.DataFrame([
    ["Data loading", "ass4_utils -> NumPy (N,C,H,W)", "same arrays, transposed t",
    ["Preprocessing", "ass4_utils (shared)", "shared",
    ["Model definition", "S.Sequential(...)", "keras.Sequential(...)",
    ["C1 / C3 conv", "S.Conv2D(k=5) im2col+matmul", "keras.layers.Conv2D(5)",
    ["Activation", "S.ReLU (mask multiply)", "activation='relu'",
    ["S2 / S4 pooling", "S.MaxPool2D (argmax routing)", "keras.layers.MaxPooling2D",
    ["Flatten", "S.Flatten (reshape)", "keras.layers.Flatten",
    ["F5 / F6 / output", "S.Dense (x@W+b)", "keras.layers.Dense",
    ["Loss", "S.SoftmaxCrossEntropy", "SparseCategoricalCrossent",
    ["Gradient", "hand-derived backward + col2im", "automatic (GradientTape)",
    ["Optimizer", "S.Adam (written out)", "keras.optimizers.Adam",
    ["Training loop", "explicit for-loop", "model.fit()",
    ["Evaluation", "predict_classes + sklearn", "predict + sklearn",
    ["Hardware", "CPU (NumPy)", "CPU (no GPU TF on Windows",
    ], columns=["Component", "Scratch", "TensorFlow/Keras", "PyTorch"])

display(component_table)
```

	Component	Scratch	TensorFlow/Keras	PyTorch
0	Data loading	ass4_utils -> NumPy (N,C,H,W)	same arrays, transposed to (N,H,W,C)	same arrays
1	Preprocessing	ass4_utils (shared)	shared	shared
2	Model definition	S.Sequential([...])	keras.Sequential([...])	nn.Module (LeNet5)
3	C1 / C3 conv	S.Conv2D(k=5) im2col+matmul	keras.layers.Conv2D(5)	nn.Conv2d(5)
4	Activation	S.ReLU (mask multiply)	activation='relu'	nn.ReLU
5	S2 / S4 pooling	S.MaxPool2D (argmax routing)	keras.layers.MaxPooling2D	nn.MaxPool2d
6	Flatten	S.Flatten (reshape)	keras.layers.Flatten	nn.Flatten
7	F5 / F6 / output	S.Dense (x@W+b)	keras.layers.Dense	nn.Linear
8	Loss	S.SoftmaxCrossEntropy	SparseCategoricalCrossentropy	nn.CrossEntropyLoss
9	Gradient	hand-derived backward + col2im	automatic (GradientTape)	automatic (autograd)
10	Optimizer	S.Adam (written out)	keras.optimizers.Adam	torch.optim.Adam
11	Training loop	explicit for-loop	model.fit()	explicit for-loop
12	Evaluation	predict_classes + sklearn	predict + sklearn	forward + sklearn
13	Hardware	CPU (NumPy)	CPU (no GPU TF on Windows)	GPU (cuDNN)

```
In [26]: os.makedirs("results", exist_ok=True)
with open("results/06_mnist_lenet.json", "w") as f:
    json.dump({"meta": {k: v for k, v in meta.items()},
              "results": [r.__dict__ for r in results],
              "full_results": [r.__dict__ for r in full_results],
              "component_table": component_table.to_dict(orient="records"),
              "baseline_comparison": pivot.reset_index().to_dict(orient="records")},
              f, indent=2, default=str)
print("saved results/06_mnist_lenet.json")
```

saved results/06_mnist_lenet.json

Saving the trained models

The JSON above records what the models *scored*. It does not record what they *are* — rerunning any of these numbers would mean retraining from scratch.

`U.save_model` writes each leg in its framework's native format, under `results/models/<notebook>/`, with a JSON sidecar naming the architecture and the

score the weights actually produced:

leg	file	reloads alone?
Scratch (NumPy)	<code>.npz</code> , one array per layer parameter	no
TensorFlow/Keras	<code>.keras</code> , graph + weights	yes
PyTorch	<code>.pt</code> , <code>state_dict</code>	no

`.pt` and `.npz` hold weights only, so reloading them means rebuilding the architecture first and passing the fresh instance to `U.load_model(..., model=...)`. That is the same contract `variants.py` uses for `results/variant_cache/`.

```
In [27]: # Weights are the model; metrics only describe it. Persist all three legs.
TAG = "06_mnist_lenet"

by_fw = {r.framework: r for r in results}

for obj, fw, stem in [
    (scratch_model, "Scratch (NumPy)", "lenet5_scratch"),
    (keras_model, "TensorFlow/Keras", "lenet5_keras"),
    (torch_model, "PyTorch", "lenet5_torch"),
]:
    r = by_fw[fw]
    U.save_model(obj, f"{stem}_subset", TAG, architecture=r.model, dataset=r.dataset,
                 epochs=r.epochs, test_accuracy=r.test_accuracy, f1_macro=r.f1_macro)

# The full-data reruns – scratch NumPy is not rerun at this size, so two legs.
by_fw_full = {r.framework: r for r in full_results}

for obj, fw, stem in [
    (keras_full, "TensorFlow/Keras", "lenet5_keras"),
    (torch_full, "PyTorch", "lenet5_torch"),
]:
    r = by_fw_full[fw]
    U.save_model(obj, f"{stem}_full", TAG, architecture=r.model, dataset=r.dataset,
                 epochs=r.epochs, test_accuracy=r.test_accuracy, f1_macro=r.f1_macro)

display(U.list_models(TAG)[["name", "framework", "n_params", "dataset", "test_accu"]])
```

```
saved results\models\06_mnist_lenet\lenet5_scratch_subset.npz (61,706 params, 226 K
B)
saved results\models\06_mnist_lenet\lenet5_keras_subset.keras (61,706 params, 765 K
B)
saved results\models\06_mnist_lenet\lenet5_torch_subset.pt (61,706 params, 245 KB)
saved results\models\06_mnist_lenet\lenet5_keras_full.keras (61,706 params, 765 KB)
saved results\models\06_mnist_lenet\lenet5_torch_full.pt (61,706 params, 245 KB)
```

	name	framework	n_params	dataset	test_accuracy
0	lenet5_keras_full	keras	61706	MNIST (full 60k)	0.9870
1	lenet5_keras_subset	keras	61706	MNIST (10k subset)	0.9765
2	lenet5_scratch_subset	scratch	61706	MNIST (10k subset)	0.9750
3	lenet5_torch_full	torch	61706	MNIST (full 60k)	0.9868
4	lenet5_torch_subset	torch	61706	MNIST (10k subset)	0.9785

10. What this notebook shows

LeNet-5 is on home ground here. Handwritten digits at 28×28 are precisely the problem it was designed around, and six edge-detecting kernels followed by sixteen composite ones are enough to separate ten digit classes.

Read the section 8 table before drawing conclusions — the delta is what matters, not the headline accuracy. Both architectures sit high enough on MNIST that the gap between them is small, which is itself the finding: **on an easy dataset, architecture choice barely registers**. The interesting comparison is in `07_cifar10_lenet.ipynb`, where the same network meets a dataset that will punish its small capacity.

One caveat worth stating: the three frameworks do not produce byte-identical numbers even with a shared seed. Initialization order and shuffle sequences differ internally, so a spread of a few tenths of a point between Scratch, Keras and PyTorch is expected noise, not evidence that one is better.